

Phase 4 - README

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PHASE 4 — Predictive & Advanced Analytics

Real Estate AI Decision Intelligence Platform

Yeh Phase 4 hai — Master A-Z Prompt ki 10-phase architecture (P1→P10) ka agla step, jo diye gaye Phase 1 (Data Foundation), Phase 2 (SQL BI), aur Phase 3 (EDA & Statistics) deliverables ke upar bana hai.

Kya banaya gaya

7 candidate predictive models (Section 4.2 ke A-G), poore ML governance ke saath (leakage checks, temporal/holdout split, baseline comparison, model registry):

Model	File	Target	Final Status
A — Property Valuation	src/model_a_valuation.py	asking_price	PASS WITH LIMITATIONS
B — Rent Prediction	src/model_b_rent.py	monthly_rent	PASS WITH LIMITATIONS
C — Price Forecast (6m)	src/model_c_price_forecast.py	locality average_price_sqft , +6m	CONDITIONAL (forced)
D — Demand Forecast (6m)	src/model_d_demand_forecast.py	locality demand_index , +6m	PASS WITH LIMITATIONS
E — Days on Market	src/model_e_dom.py	days_on_market	REJECTED
F — Sale Probability	src/model_f_sale_probability.py	Sold vs Expired/Withdrawn	REJECTED
G — Risk Score	src/model_g_risk_score.py	composite 0-100 risk score	UNCALIBRATED

Poori detail ke liye padhein: `docs/phase4_report.md` (phase completion report), `docs/model_registry.md` (har model ka full card), `docs/limitations.md` (sab limitations ek jagah).

Kaise chalayein

```
cd phase4
pip install -r requirements-phase4.txt --break-system-packages

# Sab 7 models ek saath train/compute karo (idempotent – dobara chalane se overwrite hoga, koi error
nahi)
python3 src/train_all.py

# Model registry ko readable markdown me render karo
python3 src/render_registry_md.py
```

```
# 14 automated tests chalao
python3 src/test_phase4.py
```

Data source

Yeh phase Phase 1 ke `data/processed/*.csv` files (already cleaned, validated data) se seedha padhta hai — `data/` folder me copy kiye gaye hain. Agar live PostgreSQL database available ho, to `src/data_loader.py` ko usse point kiya ja sakta hai bina kisi model script ko change kiye (join logic same rahegi).

Zaroori governance decisions (summary)

1. **Model A aur B ek doosre ka target feature ke roop me use nahi karte** (`asking_price` ↔ `monthly_rent`), taaki circular leakage na ho — automated test se enforce kiya gaya hai.
2. **Model C ka status CONDITIONAL force kiya gaya hai**, uske metrics achhe hone ke bawajood, kyunki uska target column Phase 3 EDA me traced ek known data-generation bug (uncapped compounding) carry karta hai. Original generator script deliverables me nahi tha, isliye source par fix nahi ho saka — documented log-space fallback use kiya gaya.
3. **Model E aur F ko REJECT kiya gaya** kyunki independently verify karne par unka signal near-random nikla (yeh Phase 3 EDA ke findings ko confirm karta hai).
4. **Model D baseline (persistence) ko beat nahi kar paya**, isliye high R^2 ke bawajood usko PASS nahi diya gaya.
5. **Model G ek transparent, deterministic composite hai** (trained ML model nahi), REJECTED models (E, F) ko input ke roop me exclude karta hai, aur UNCALIBRATED status ke saath register hua hai kyunki real-world outcome se validate nahi kiya gaya.

Folder structure

```
phase4/
├─ data/           # Phase 1 se copy kiya gaya processed data (read-only input)
├─ src/            # sab model scripts, data loader, governance utils, tests
├─ models/         # trained .joblib artifacts + model_registry.json
├─ docs/           # phase4_report.md, model_registry.md, limitations.md
├─ requirements-phase4.txt
└─ README_PHASE4.md # yeh file
```

Next phase dependency

Phase 5 (Evidence Intelligence & RAG) is phase par directly depend nahi karta. Phases 6/7/8 Model A/B/C/D/G ke outputs consume karenge jaisa `docs/phase4_report.md` Section 12 me likha hai. Phase 9 ka Prediction Agent `models/model_registry.json` load karke REJECTED models ko serve karne se mana karega.